

# Atomic Design System Manual

## Pangasinan Heritage Digital Showcase — Activity 1.2

**Methodology:** Brad Frost's Atomic Design

**Framework:** Next.js 14 (App Router) + TypeScript + Tailwind CSS

**Date:** August 31, 2026

### Overview

This manual documents the modular component library for the Pangasinan Heritage Digital Showcase. Components are organized into three Atomic Design levels:

| Level     | Components                                    |
|-----------|-----------------------------------------------|
| Atoms     | Button, Typography, Color Tokens, Icon, Image |
| Molecules | Heritage Card, Search Form, Navigation Item   |
| Organisms | Heritage Grid, Header Navigation              |

Each component section includes: Visual Preview, Usage Context, Responsive Logic, and Code Reference.

### Atoms

#### Button

**Visual Preview:** Primary (ocean blue), Secondary (sand gold), Outline, and Ghost variants. Minimum height 44px for touch accessibility.

**Usage Context:** Primary call-to-action control for forms, navigation prompts, and heritage site exploration links. Used in the Search Form submit action and the "View Design System" footer CTA.

#### Responsive Logic:

- Minimum touch target: 44px ( min-h-11 ) on all breakpoints
- Sizes: sm (36px), md (44px), lg (48px)
- fullWidth prop stretches button to 100% container width on mobile forms
- Focus ring visible at all sizes for keyboard navigation (WCAG 2.4.7)

#### Code Reference:

```
import { Button } from "@/components/atoms/Button/Button";

<Button variant="primary" size="md">Explore site</Button>
<Button variant="outline" fullWidth>Search</Button>
```

#### Typography

**Visual Preview:** Seven variants — display, h1, h2, h3, body, body-sm, caption, label — with four color modes (default, muted, inverse, accent).

**Usage Context:** Semantic text styles for headings, body copy, labels, and captions across all heritage pages. Ensures consistent typographic hierarchy and readable line lengths.

**Responsive Logic:**

- `display` : text-3xl (mobile) → text-4xl (sm) → text-5xl (lg)
- `h1` : text-2xl → text-3xl at sm:
- `h2` : text-xl → text-2xl at sm:
- Body text maintains 16px base size (WCAG minimum readable size)
- `as` prop allows semantic override (e.g., `variant="h2" as="h1"` )

**Code Reference:**

```
import { Typography } from "@components/atoms/Typography/Typography";

<Typography variant="display">Discover Pangasinan's Heritage</Typography>
<Typography variant="body" color="muted">{description}</Typography>
```

**Color Tokens**

**Visual Preview:** Swatches for ocean (coastal), sand (cultural warmth), forest (natural), and neutral palettes, plus semantic error/success colors.

**Usage Context:** Centralized color palette reflecting Pangasinan's coastal, cultural, and natural identity. Applied via Tailwind CSS custom theme extension and TypeScript token exports.

**Responsive Logic:**

- Documentation grid: 1 column (mobile) → 2 columns (sm) → 3 columns (lg)
- Color values remain constant across breakpoints; contrast ratios verified for WCAG AA (4.5:1 text, 3:1 UI)

**Token Values:**

| Token       | Hex     | Usage                          |
|-------------|---------|--------------------------------|
| ocean-500   | #1B4B7A | Primary brand, links, headers  |
| sand-500    | #C9A227 | Accent, secondary buttons      |
| forest-500  | #2D5A27 | Nature/eco heritage categories |
| neutral-900 | #111827 | Body text                      |
| neutral-500 | #6B7280 | Muted/secondary text           |

**Code Reference:**

```
import { colorTokens } from "@tokens/colors";

// Access: colorTokens.ocean[500] → "#1B4B7A"
// Tailwind: className="bg-ocean-500 text-white"
```

**Icon**

**Visual Preview:** Inline SVG icons — search, menu, close, location, chevron-right, heritage — rendered at configurable sizes without external font dependencies.

**Usage Context:** Navigation indicators, search field decoration, location markers on Heritage Cards, and brand logo in the header. Zero network requests (no icon font CDN).

**Responsive Logic:**

- Default size: 20px; accepts `size` prop for scaling
- Icons inside buttons inherit `currentColor` for theme consistency
- `label` prop enables accessible name when icon is standalone interactive element

**Code Reference:**

```
import { Icon } from "@components/atoms/Icon/Icon";

<Icon name="location" size={16} className="text-ocean-500" />
<Icon name="menu" size={22} label="Open menu" />
```

## Image (HeritageImage)

**Visual Preview:** Responsive image container with locked aspect ratios (16:9, 4:3, 1:1), rounded corners, and lazy loading.

**Usage Context:** Optimized heritage imagery wrapper using Next.js `Image` for automatic lazy loading, responsive `srcset`, and WebP conversion — critical for users on limited mobile data.

**Responsive Logic:**

- `sizes` attribute: `(max-width: 640px) 100vw, (max-width: 1024px) 50vw, 33vw`
- Aspect ratio locked via CSS `aspect-ratio` to prevent layout shift (CLS)
- Images lazy-loaded below the fold; hero images use `priority` prop
- Container rounds corners at `rounded-xl` (12px)

**Code Reference:**

```
import { HeritageImage } from "@components/atoms/Image/HeritageImage";

<HeritageImage
  src="https://images.unsplash.com/photo-1507525428034-b723cf961d3e"
  alt="Hundred Islands National Park in Alaminos, Pangasinan"
  aspectRatio="16/9"
/>
```

## Molecules

### Heritage Card

**Visual Preview:** Card with heritage site image, category badge, title, location pin, truncated description, and "Explore site" link.

**Usage Context:** The Heritage Card is used exclusively for displaying tourist site previews in a responsive heritage-site grid. Each card represents one Pangasinan destination (e.g., Hundred Islands, Bolinao Lighthouse).

**Responsive Logic:**

- Full-width within grid cell on all breakpoints
- Image: 16:9 aspect ratio, full card width
- Description: `line-clamp-3` truncates to 3 lines on all sizes
- Padding: `p-4` (mobile) → `p-5` (sm+)
- Hover: subtle shadow elevation ( `hover:shadow-md` )
- Card height: `h-full` with flex column ensures equal heights in grid rows

**Code Reference:**

```
import { HeritageCard } from "@components/molecules/HeritageCard/HeritageCard";

<HeritageCard site={heritageSite} href="/sites/hundred-islands" />
```

## Search Form

**Visual Preview:** Search input with magnifying glass icon and submit button. Labeled for screen readers.

**Usage Context:** Site-wide heritage search in the hero section. Filters sites by name, location, category, and description on the home page.

**Responsive Logic:**

- Mobile: stacked vertically ( `flex-col` ) — input full width, button full width
- Tablet+: horizontal row ( `sm:flex-row` ) — input flex-1, button fixed min-width 120px
- Input min-height: 44px for touch targets
- Focus ring on input: `ring-ocean-500/30` for visible focus indicator

**Code Reference:**

```
import { SearchForm } from "@components/molecules/SearchForm/SearchForm";

<SearchForm onSearch={(query) => filterSites(query)} />
```

## Navigation Item

**Visual Preview:** Text link with optional icon, active state highlighting (ocean-50 background), and hover effects.

**Usage Context:** Individual navigation link used inside the Header Navigation organism for primary site sections (Home, Heritage Sites, Design System).

**Responsive Logic:**

- Minimum tap target: 44px height ( `min-h-11` )
- Active state: `bg-ocean-50 text-ocean-700` with `aria-current="page"`
- Hover: `bg-neutral-100` on inactive items
- Used in horizontal row (desktop) and vertical stack (mobile menu)

**Code Reference:**

```
import { NavigationItem } from "@components/molecules/NavigationItem/NavigationItem";

<NavigationItem label="Heritage Sites" href="/#sites" isActive={false} />
```

## Organisms

### Heritage Grid

**Visual Preview:** Section heading, description, and responsive grid of Heritage Cards. Shows empty state when no results match search.

**Usage Context:** Main listing organism on the home page. Composes Heritage Card molecules into a filterable, responsive grid of all Pangasinan heritage destinations.

#### Responsive Logic:

- Grid columns: 1 (default) → 2 ( sm:grid-cols-2 ) → 3 ( lg:grid-cols-3 )
- Gap: gap-5 (mobile) → gap-6 (sm+)
- Section heading and description above grid with mb-6 (mobile) → mb-8 (sm+)
- Empty state: text message with role="status" for screen reader announcement

#### Code Reference:

```
import { HeritageGrid } from "@components/organisms/HeritageGrid/HeritageGrid";

<HeritageGrid
  sites={filteredSites}
  title="Featured Heritage Sites"
  description="Discover Pangasinan's most treasured destinations."
/>
```

### Header Navigation

**Visual Preview:** Sticky header with brand logo, desktop horizontal nav, and mobile hamburger menu with slide-down panel.

**Usage Context:** Persistent site header on all pages. Provides primary navigation, brand identity, and mobile-friendly menu toggle.

#### Responsive Logic:

- Sticky: sticky top-0 z-50 with backdrop blur
- Desktop nav: visible at md: breakpoint and above
- Mobile: hamburger button toggles collapsible menu below header
- Menu button: aria-expanded and aria-controls for accessibility
- Mobile menu: full-width vertical stack of Navigation Items
- Brand: icon + text, scales from text-base to text-lg at sm:

#### Code Reference:

```
import { HeaderNavigation } from "@components/organisms/HeaderNavigation/HeaderNavigation";

<HeaderNavigation currentPath="/" />
```

## File Structure

```
src/
├─ tokens/
│   └─ colors.ts
├─ components/
│   └─ atoms/
│       ├── Button/
│       ├── Typography/
│       ├── ColorTokens/
│       ├── Icon/
│       └─ Image/
│   └─ molecules/
│       ├── HeritageCard/
│       ├── SearchForm/
│       └─ NavigationItem/
│   └─ organisms/
│       ├── HeritageGrid/
│       └─ HeaderNavigation/
├─ data/
│   └─ heritage-sites.ts
└─ app/
    ├── page.tsx           (Home – uses all organisms)
    └─ design-system/
        └─ page.tsx       (Live component documentation)
```

## Accessibility Checklist (WCAG 2.1 AA)

| Requirement                  | Implementation                                               |
|------------------------------|--------------------------------------------------------------|
| 1.4.3 Contrast (Minimum)     | ocean-500 on white: 7.2:1; neutral-900 on white: 15.8:1      |
| 2.4.7 Focus Visible          | focus-visible:ring-2 on all interactive elements             |
| 2.5.5 Target Size            | Minimum 44×44px touch targets on buttons, links, inputs      |
| 4.1.2 Name, Role, Value      | aria-label, aria-current, role="search" applied              |
| 1.3.1 Info and Relationships | Semantic HTML: <header>, <nav>, <main>, <article>, <section> |